
CS-4096 — Data Science for Software Engineering

Lecture Notes: Data Types and Correlation Statistics

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Types of Data in Data Science

Data can be broadly categorized based on structure, measurement scale, and numerical properties.

1. Categorical (Qualitative) Data

Non-numeric data representing labels or groups. Arithmetic operations are not meaningful.

(a) Nominal Data

Definition: Categories with no natural order or ranking.

Properties: Only equality comparison allowed ($=$, $\neq$).

Examples:

- Gender: Male, Female
- Blood Group: A, B, AB, O
- Country names

(b) Ordinal Data

Definition: Categories with a meaningful order but unequal differences between levels.

Properties: Ranking possible, arithmetic not meaningful.

Examples:

- Satisfaction: Low, Medium, High
- Grades: A, B, C, D
- Education level: BS < MS < PhD

2. Numerical (Quantitative) Data

Numeric values where arithmetic operations are meaningful.

(a) Discrete Data

Definition: Countable integers obtained by counting.

Properties: No intermediate values.

Examples:

- Number of students in class
- Number of bugs in software
- Number of purchases

(b) Continuous Data

Definition: Measured values that can take any value within a range.

Properties: Infinite possible values (decimals allowed).

Examples:

- Height, weight
- Temperature
- Time, speed

3. Measurement Scale Types

(a) Interval Scale

Definition: Equal intervals between values but no true zero.

Properties: Addition/subtraction valid, ratios are not meaningful.

Examples:

- Temperature in °C or °F
- Calendar years

(b) Ratio Scale

Definition: Equal intervals with a true zero point.

Properties: All arithmetic operations valid (including ratios).

Examples:

- Income
- Distance
- Age
- Length/weight

Summary Table

Type	Ordered	Arithmetic	Example
Nominal	No	No	Gender
Ordinal	Yes	No	Grades
Discrete	Yes	Yes	Bug count
Continuous	Yes	Yes	Height
Interval	Yes	+,- only	Temperature
Ratio	Yes	All ops	Salary

Quick Rule of Thumb

- Labels $\rightarrow$ Nominal
- Ranks $\rightarrow$ Ordinal
- Counts $\rightarrow$ Discrete
- Measurements $\rightarrow$ Continuous
- No true zero $\rightarrow$ Interval
- True zero $\rightarrow$ Ratio

Part II: Correlation Measures Based on Data Types

Correlation quantifies the strength and direction of association between variables. The choice of correlation method depends on the **measurement scale** of the variables.

1. Numerical vs Numerical

(a) Pearson Correlation (r)

Use when: Both variables are continuous (interval/ratio) and normally distributed.

Measures: Linear relationship.

Range: $-1 \leq r \leq 1$

Formula:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Examples:

- Height vs Weight
- Study hours vs Marks

(b) Spearman Rank Correlation (ρ)

Use when: Ordinal OR non-normal continuous data.

Measures: Monotonic relationship using ranks.

Formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Examples:

- Class rank vs Performance rank
- Satisfaction level vs Service quality

(c) Kendall's Tau (τ)

Use when: Small datasets or many tied ranks.

Measures: Concordant vs discordant pairs.

Examples:

- Preference rankings

2. Categorical vs Categorical

(a) Chi-Square Association Test

Use when: Both variables are nominal.

Measures: Independence between categories.

Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Examples:

- Gender vs Product choice
- Department vs Pass/Fail

(b) Cramér's V

Use when: Strength of association for nominal variables.

Range: 0 to 1

Formula:

$$V = \sqrt{\frac{\chi^2}{n(k - 1)}}$$

3. Numerical vs Binary/Categorical

(a) Point-Biserial Correlation

Use when: One continuous + one binary variable.

Examples:

- Gender vs Salary
- Treatment (Yes/No) vs Recovery time

(b) ANOVA / Eta (η) Coefficient

Use when: One continuous + one multi-class categorical.

Measures: Group mean differences.

Range: 0 to 1

4. Binary vs Binary

(a) Phi Coefficient (ϕ)

Use when: Two binary variables.

Range: -1 to 1

Example:

- Smoker (Yes/No) vs Disease (Yes/No)

5. Quick Selection Guide

Variable Type 1	Variable Type 2	Recommended Measure
Continuous	Continuous	Pearson
Ordinal	Ordinal	Spearman / Kendall
Nominal	Nominal	Chi-square / Cramér's V
Continuous	Binary	Point-biserial
Continuous	Categorical (multi)	ANOVA / Eta
Binary	Binary	Phi

Practical Rule of Thumb

- Linear numeric $\rightarrow$ Pearson
- Ranked data $\rightarrow$ Spearman
- Categories $\rightarrow$ Chi-square
- Binary + numeric $\rightarrow$ Point-biserial

- Two binaries $\rightarrow$ Phi

THE END